10220

Aqua Green

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A4

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Abstract

Egypt faces several major challenges, including Population growth and its consequences, and addressing and reducing pollution that fouls our air, water, and soil. Egypt suffers from increasing food demand due to population growth, which places pressure on agricultural systems that rely on limited freshwater resources and unsustainable farming practices. The project aims to solve these problems by creating hybrid aquaponics. The prototype features a 0.5 m^2 greenhouse integrated with a 10 L aquaculture tank, creating a closed-loop nutrient cycle. This project will be portable and testable under different conditions, so the system was specifically engineered for the high-humidity environment (avg. 70%) of Borg El Arab City, supporting Egypt's goal of expanding its industrial and agricultural base, furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egyptian Grand Challenge of dealing with population growth and its consequences. The project aims to measure three parameters (Temperature, light intensity, and water level), and they are treated by four actuators (water pump, LED, heater, and DC fan). The project includes a preloaded database with optimized environmental profiles for plant species (Lettuce and Basil) and aquatic organisms (Tilapia and Salmon). The prototype has a positive result like The prototype recorded an average temperature of **20.34°C**, with a minimum of **18.31°C** and a maximum of **36.75°C**, showing generally stable conditions with occasional increases. The light intensity (lux) had a mean of **827.39 lux**, ranging from a minimum of **31 lux** to a maximum of **5884 lux**, indicating wide variation in lighting levels. Additionally, the ultrasonic sensor measured distance with an average of **17.98 cm**, a minimum of **5.20 cm**, and a maximum of **39.98 cm**, reflecting changes in water level during system operation and some negative results such as fish death and some electronic damage.

Introduction

Egypt is currently facing several grand challenges affecting the food demand and freshwater resources, as well as unsustainable farming practices. One of them is population growth, **which is shown in Figure 3**, and its consequences. An increase in population growth places a heavy load on the Egyptian economy, and it influences the food demand. Also, addressing and reducing pollution fouling our air, water, and soil is one of these challenges. Reports say that 38 million Egyptians drink water polluted by sewage, leading to the spread of diseases such as cholera and hepatitis, which negatively affect the limited freshwater resources and unsustainable farming practices.

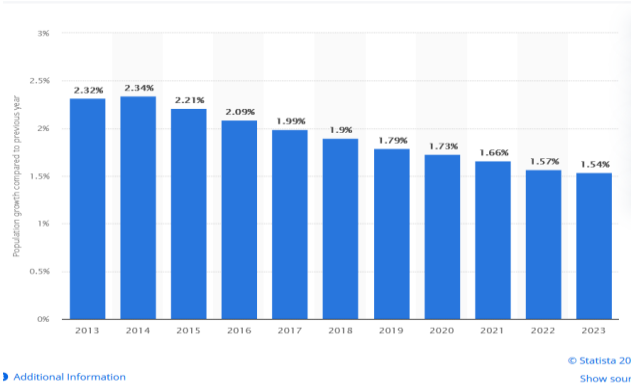


Figure 1 shows the population growth percentage, Statista. (2024, September 27).

Therefore, in this semester, we were asked to design and implement an automated closed-loop control system to monitor and regulate essential environmental variables in aquaponics. So, our prototype focused on a specific problem, food scarcity, and traditional irrigation.

In researching the Problem to Be Solved, we analyzed prior solutions like Upward Farms and Aquaponics Farm Al-Bustan. These systems demonstrate a major strength in water conservation, using recycling to save up to 90% more water than traditional farming. However, a significant weakness is the high operational cost of artificially recreating sunlight and airflow. Upward Farms had to pay to artificially recreate all those things 24 hours a day, 7 days a week.

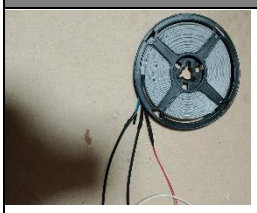
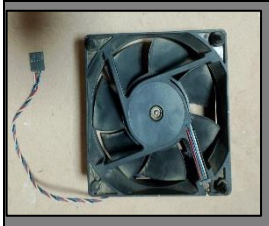
To meet the demands of a growing population and rising water scarcity, agriculture must transition from traditional irrigation to more efficient, targeted methods. Our team chose a hybrid aquaponics solution that utilizes a symbiotic cycle: fish waste provides nutrients for plants, while plants naturally purify the water for the fish. We chose this specific design because it offers a portable, cost-effective, and closed-loop ecosystem that minimizes waste. This harmonious relationship between aquatic life and vegetation fosters a closed-loop ecosystem, significantly curbing water wastage and elevating the system's overall sustainability.

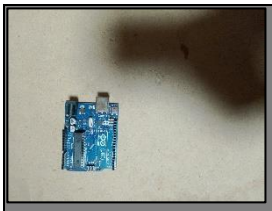
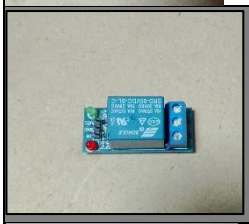
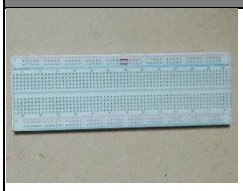
To ensure success, our prototype had to meet strict Design Requirements: a 0.5 m^2 footprint, a 10 L aquaculture tank, and an automated system to manage three

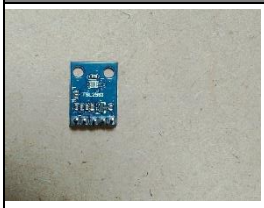
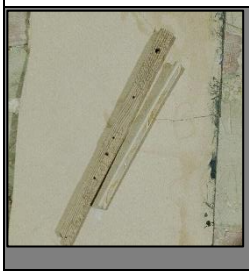
parameters (temperature, light intensity, and water level). These are regulated via setpoints in an Arduino-based controller that triggers a water pump, LED array, heater, and DC fan. Additionally, the system features a preloaded database for Lettuce, Basil, Tilapia, and Salmon to automate environmental profiles. By utilizing recycled wood, plastic, and glass, we ensured the prototype is both sustainable and accessible. To achieve this functional design, specific Materials and Methods were employed to construct and rigorously test the system's performance.

Materials & methods

Materials

Picture	Name	Quantity	Description	Price
	Heater	1	Raising the water temperature of the fish tank	475 EGP
	LED	1	The responsible part for the plant's photosynthesis	147 EGP
	DC fan	1	Reducing the fish water temperature	recycled
	Servo motor	1	Control of the feeding system	75 EGP

	Arduino	1	The brain of the circuit	Recycled
	LCD	1	Display sensors readings	Recycled
	Relay	3	control on the circuit by an independent low-power signal	45EGP
	Breadboard	2	Connection of the circuit	Recycled
	Power supply	1	Circuit Power source	290 EGP
	Temperature sensor	1	Read the water temperature	Recycled
	Water tube	3	Moving of the cycle water	Recycled
	Fish tank	1	The fish's home	Recycled

	PVC pipe	1	Carry the plants and the body of the greenhouse	Recycled
	Light sensor	1	Read the light intensity	190 EGP
	Water bottle	1	Work as a sump	Recycled
	Water pump	2	Push the water between the greenhouse and the fish tank	290 EGP
	Pieces of wood	4	Carry the prototype and cover the greenhouse	Recycled

Methods

To build the prototype, you should follow these steps.

Figure 2 shows while cutting the green house.

1-We used a recycled wooden box to carry the prototype as a frame; the frame has a height of 130 cm, a length of 70 cm, and a width of 50 cm.

2-We cut pieces of wood for a greenhouse, which has an isosceles triangle with each side measuring 40 cm, 15 cm height, and 50 cm width, **as shown in Figure 2**, fixed on the wood box, also covered with a plastic cover, and built a greenhouse gate.

Figure 3 shows while painting the frame

3-We painted the wooden box a brown colour and the greenhouse a white colour, **as shown in Figure 3.**

4 Cut a recycled PVC pipe and make three equal holes on the top of it to put the plants in these holes.

Figure 4 shows while washing the fish tank

4-We washed a recycled fish tank, **as shown in Figure 4**, and put in beneficial bacteria and Chlorine remover to make the water ready to put the fish in it, and added the heater to the fish tank. Also, fixing the servo motor and the DC fan above the fish tank.

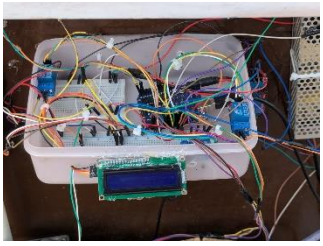


Figure 5 shows the circuit.

5-Connect the Arduino with sensors: a light sensor, a temperature sensor, and a water level sensor, and actuators: DC fan, pumps, LED, Heater, and Servo motor, **as shown in Figure 5.**

6-Cutting the water tube into three parts, one reaches from the fish tank to the PVC pipe, the second one from the PVC to the bottle of water, and the final one from the bottle to the fish tank.

7-Put two pumps, one from the fish tank into the greenhouse, and the second one from the bottle to the fish tank.

8-Make a biofilter with cotton, gravel, and clay, and put it in a small water bottle, **as shown in Figure 6.**

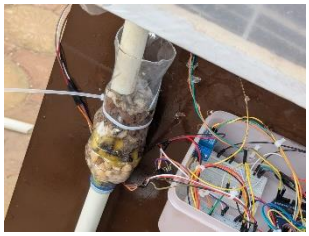


Figure 6 shows the biofilter.

Calibration

Figure 7: Calibration curve of the ultrasonic sensor comparing measured distance to real distance. The red calibration line (y = x - 1.5) indicates a consistent negative offset of 1.5 cm compared to the dashed blue ideal line (y = x).

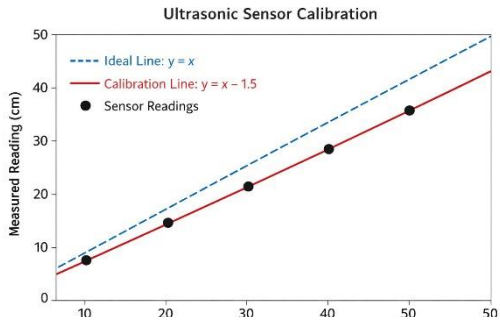


Figure 7 shows the calibration of the Ultrasonic sensor

Figure 8 shows the calibration of the light sensor

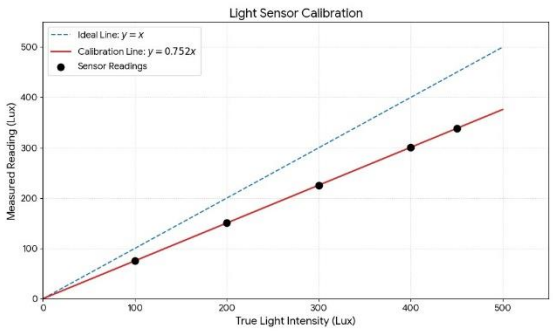


Figure 8 shows the calibration of the light sensor

Test plan

After constructing the prototype, it should be tested to ensure its efficiency.

First, the body of the prototype was tested by forcing it with a high weight to ensure its stability. Before any electronic connection, the sensor's calibration was done by scientific procedures. An electric circuit test was done through three stages: Firstly, ensure all three sensors are connected and take down their readings with the LCD screen.

The second step was testing all actuators with an if statement and a while loop and writing down the time taken to apply system recovery.

The third step was connecting all sensors and actuators with each other, and ensuring they work in concert to achieve maximum efficiency. The upper pipe was tested to determine if it dropped water.

After these tests, the prototype is professional and precise.

Design requirements

The chosen solution was an aquaponic system. It is an automated system with a closed-loop system. Thus, before designing the prototype, setting requirements is needed to define its success and the results intended to be achieved. Most of the prototype's body was built from recycled materials, such as PVC pipe, a fish tank, a wood frame, and some recycled electronics, like an Arduino, an LCD screen, wires, DC fan. The fish tank is 10 L, and the greenhouse is less than .05 m^2, which achieves the design requirements. The system treats three parameters (water level, temperature, and light intensity), and contains four actuators, a DC fan, a heater, an LED, and pumps. Each parameter has been calibrated with a real device. The system has the order of (IF, then), exactly in the pump. If the water level in the fish tank decreases by a certain distance, the pump will turn on, and we follow all the design requirements to achieve all requirements.

Results

Negative Results

Dead fish

Before completing the prototype, fish were introduced into the tank under basic conditions, including feeding, a heater, and an air pump. However, after three days, some fish began to die.

After observation and analysis, it was concluded that the primary cause was insufficient oxygen dissolved in the water. In addition, the absence of an efficient filtration system led to poor water quality.

This resulted in the accumulation of waste materials, which likely increased ammonia levels and further stressed the fish. It was also observed that the water required filtration at least twice per week to maintain suitable living conditions.

Lettuce Wilting

Before starting the prototype, lettuce plants without fully developed roots were purchased and placed in water to stimulate root growth. However, after three days, signs of wilting were observed in some plants.

This was mainly due to excessive water exposure and the lack of proper oxygen availability around the base of the plant. Without sufficient oxygen, root development was limited, leading to stress and reduced water and nutrient uptake. As a result, the leaves began to wilt.

Damage to some electronics

Damage to some electronic components occurred during circuit testing, including an ultrasonic sensor, a power supply unit, and an LCD screen. The causes of failure were excessive input voltage applied to the ultrasonic sensor, damaged resistors within the power supply circuit, and physical damage where the LCD wires were cut.

Positive results

Table 1 shows the Power consumption analysis of the prototype components, showing the voltage, current, and power requirements for each actuator. The results indicate that the total power consumption reached **279.5 W**, with the fish tank heater and LED strip being the highest energy-consuming components, reflecting their significant role in maintaining system operation.

Table 1 shows the power consumption for each actuator

Component	Voltage (V)	Current (A)	Power (W)	Energy per Second (J/s)
Water Pump 1	12	0.5	6	6
Water Pump 2	12	0.5	6	6
LED Strip (4m)	12	4.8	57.6	57.6
Cooling Fan	12	0.2	2.4	2.4
Fish Tank Heater	220	0.9	200	50
Servo Motor (Blue)	5	0.5	2.5	2.5
Air Pump	220	0.4	5	5
Fish tank lamp	220	0.45	10	
Total			279.5	129.5

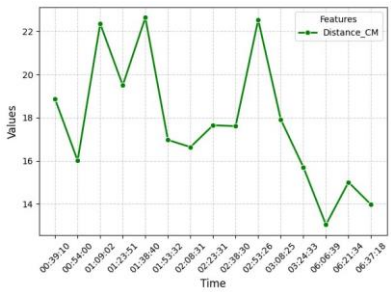


Figure 9 Ultrasonic sensor reading

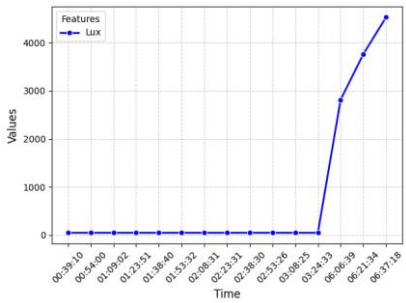


Figure 10 shows the light sensor readings

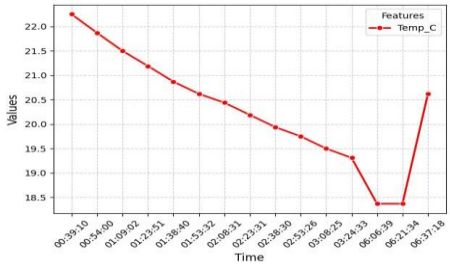


Figure11 shows the temperature sensor

Ultrasonic sensor reading:

Ultrasonic is responsible for detecting the distance between the top border of the fish tank and the water level. This parameter wasn't affected continuously, that's because decreasing water depends on the sunlight. **As shown in Figure 9**.

Light sensor reading:

This sensor was installed in the greenhouse to be exposed to light, such as sunlight. It helps the system detect the light intensity. It's observed that the deviation of the light is frequent**. As shown in Figure 10.**

Temperature sensor reading:

It's important to know any deviation in the water temperature to preserve the fish. **As shown in Figure 11.**

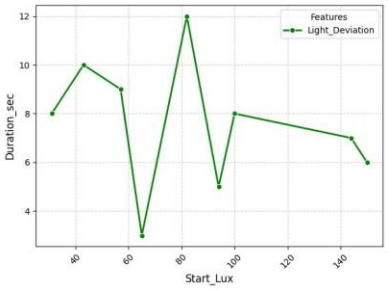


Figure 12 shows light Deviation

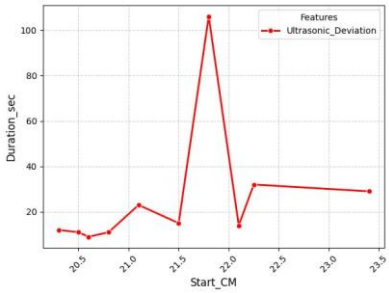


Figure 13 Ultrasonic Deviation

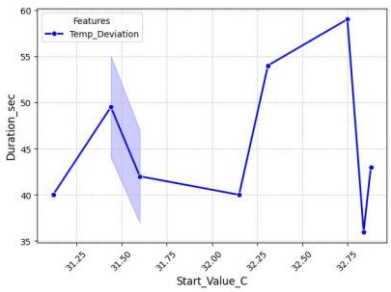


Figure 14 shows the temperature deviation

Figure 12 shows light deviation**.**

Figure 13 shows the ultrasonic Deviation.

Figure 14 shows the temperature Deviation

Analysis

Nitrogen cycle

Nitrogen is an essential element for all living organisms because it is a key component of DNA, RNA, amino acids, proteins, and other cellular structures. In aquaponic systems, the primary source of nitrogen is fish feed. After digestion, fish excrete nitrogen mainly as ammonia (about 90%).

In water, ammonia exists in two forms: un-ionized ammonia (NH_3) and ionized ammonium (NH_4^+). The total of both forms is called Total Ammonia Nitrogen (TAN). The balance between NH_3 and NH_4^+ depends on environmental conditions such as pH and temperature. Higher pH and temperature increase the proportion of NH_3, a highly toxic compound to fish.

Ammonia and nitrite are dangerous because they can damage fish gills and interfere with oxygen transport in the blood. Therefore, these compounds must be quickly converted into less harmful substances.

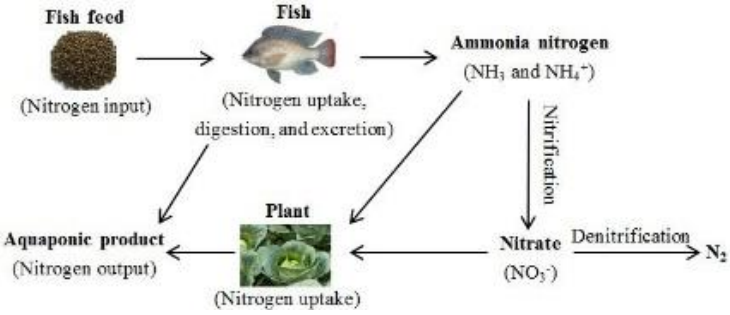


Figure 15 shows the Nitrogen cycle.

The nitrogen cycle in aquaponic systems depends on beneficial bacteria. Ammonia-oxidizing bacteria (AOB), such as *Nitrosomonas*, convert ammonium into nitrite (NO_2^-). Then, nitrite-oxidizing bacteria (NOB), such as *Nitrobacter*, convert nitrite into nitrate (NO_3^-). This process is known as nitrification, and it is **shown in Figure 15.**

Nitrate is much less toxic and is absorbed by plants as a nutrient. Plants use nitrate to produce organic nitrogen compounds such as amino acids and proteins. This process improves Nitrogen Use Efficiency (NUE), which refers to how effectively nitrogen is converted into plant biomass.

The main chemical reactions involved are:

Ammonia equilibrium in water:

$$\boldsymbol{NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-}$$

Ammonia oxidation (first nitrification step):

$$\boldsymbol{NH_4^+ + 1.5O_2 \rightarrow NO_2^- + 2H^+ + H_2O}$$

Nitrite oxidation (second nitrification step):

$$\boldsymbol{NO_2^- + 0.5O_2 \rightarrow NO_3^-}$$

Plant uptake (assimilation):

$$\boldsymbol{NO_3^-} \rightarrow \textbf{Organic Nitrogen (amino acids, proteins)}$$

The microcontroller connections

The microcontroller is called the brain of the embedded system, acting as the communication interface between sensors and actuators. To begin with, sensors collect information regarding the physical parameters from the real-world environment. Further, the collected information is processed by the microcontroller, and decisions related to the actions taken by the actuators are made based on that interpretation. **As shown in Figure 16**, the actuators are motor-based devices. An actuator is a device in the control system that assists the control process in executing its activities or actions. These devices use electric energy and convert it to mechanical movement. Four distinct stages include:

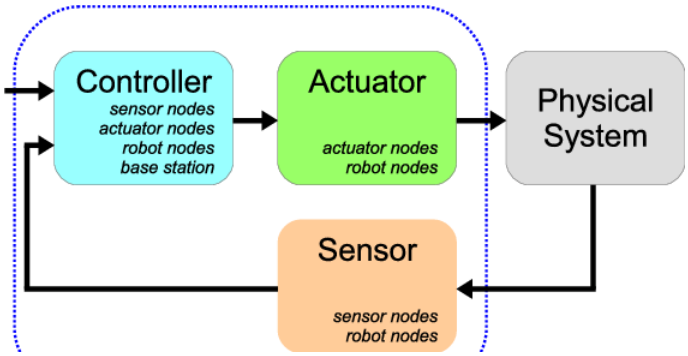


Figure 16 shows the microcontroller connections

Collection of Data through Sensors: Signals in sensors occur, and they are either analog or digital for the microcontroller to use. Analog Sensors: Produce signals in terms of voltage level, which require converting to digital through the ADC process, while Digital signals represent only two states (HIGH or LOW).

$$Vin = \frac{ADC\ value}{1023} \times Vref$$

Processing of the Signals from Sensors: This involves the microcontroller performing filtration and calibration on the signals, as well as algorithms. For example, it may compare the sensor value to a specific condition.

$$Output\ = \{\ ON\ \ if\ Sensor\ Value\ >\ Threshold$$

$$OFF\ otherwise\ \}$$

In addition, Information collected could be sent to a database using communication techniques such as wired and wireless communications. Finally, the Arduino sends output signals based on the processed data. These outputs can control devices such as LEDs, motors, or buzzers. Additionally, Arduino can transmit data to other devices or systems using communication methods such as serial communication, I2C, or SPI.

Ultrasonic Sensor

Mechanism:

Ultrasonic sensors work as eyes in living organisms. But its mechanisms of work are different. It works by sound waves at 40 kHz, which is never detected by human ears, as it is out of the human ear's allowed sound range. This sensor emits these waves to go and after hitting somebody, this sound reflects to the sensor again, **as shown in Figure 17**, and by these variables, this sensor can determine the distance according to this equation

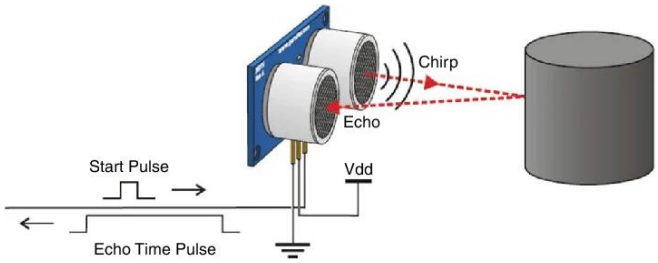


Figure 17 shows the mechanisms of the ultrasonic.

d = (v × t) / 2.

Where d is the distance to the water surface in meters, v is the speed of sound in air (approximately 343 m/s), and t is the total round-trip time in seconds. The division by 2 accounts for the fact that the sound travels to the surface and back.

Role:

This sensor is used to detect dropped below a defined minimum threshold in the water level of the fish tank, and it is connected to a water pump as an actuator, which pumps and increases the water level of the tank.

DS18B20 Temperature Sensor

Mechanism

The DS18B20 is a digital temperature sensor that measures water temperature with a resolution of up to 0.0625°C. The DS18B20 sensor converts ambient thermal energy into a digital value by measuring the frequency shift of an internal temperature-dependent oscillator. This digital data is transmitted to a microcontroller via the 1-Wire protocol, which allows bidirectional communication over a single signal line labeled IO. The pull-up resistor (R7) It is essential to this process, as it maintains the signal line at a "High" voltage state so the sensor can pulse the line "Low." to send data bits, **as shown in Figure 18**. At its core, the sensor contains a temperature-sensitive resistor whose Electrical resistance changes predictably with temperature. And it has a built-in system that converts analog to digital signals. The temperature-to-digital conversion follows this relationship for its 12-bit resolution mode: T = raw value × 0.0625 , Where raw value is the integer output from the sensor, and T is the temperature = 24°C exactly. Role: This sensor controls 2 actuators and 2 functions (heating & cooling).

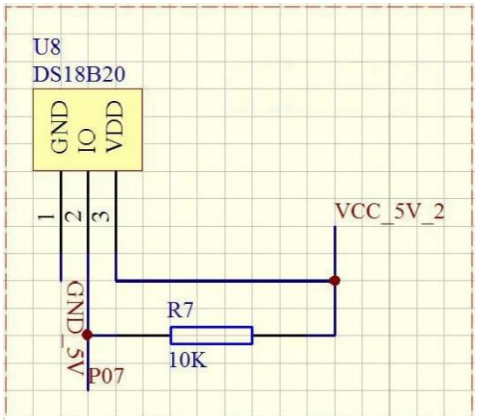


Figure 18 shows the mechanisms of the DS18B20 Temperature Sensor

If the sensor reports a temperature below 24°C, the Arduino activates the water heater to warm the tank. If the temperature exceeds 30°C, the heater is switched off, and the DC fan is activated to cool the water through evaporation and airflow across the surface.

TSL2561 Light Sensor

Mechanism:

The TSL2561 is a light-to-digital converter that measures in lux, the standard unit for visible light intensity.

It contains two photodiodes with different spectral sensitivities.

The first photodiode, called Channel 0, responds to the full visible and infrared spectrum.

The second photodiode, called Channel 1, responds primarily to infrared light only.

And the plant just takes the benefit or does its vital process according to the visible light, so the infrared we should subtract from the total light. And the sensor does that according to the following equation: Lux = **C0 × CH0 − C1 × CH1.**

Where CH0 and CH1 are the raw counts from the two photodiode channels, and C0 and C1 are calibration coefficients. This subtraction approach is what makes the TSL2561 significantly more accurate than a simple single-photodiode sensor.

Role: And this sensor is connected to the LED light, which switches it on when the lux below down about safe zone or the minimum lux required for the plant can do their vital processes.

<u>Efficiency</u>

First step: Evaluating the reading of the sensors and comparing it to reference sensors, such as the sensors in the laboratory. And calculate the error by Mean Squared Error (MSE) or by Mean Absolute Error (MAE).

MSE: $(1/n) \times \Sigma (y_i - \hat{y}_i)^2$

MAE: $(1/n) \times \Sigma |y_i - \hat{y}_i|$

Second step:

Normalization of the error by dividing the error by the maximum reading of the sensor, because the error of the sensor is in its range from 0 to 50, as a temperature sensor, it is impossible to compare it with a sensor whose range is from 0 to 40000.

Third step: Calculate the efficiency of each sensor by (100% - (normalization error)%).

Calculation of the whole system by taking the average efficiency of all sensors.

Conclusion

The project aimed to address population growth and food scarcity by developing a closed-loop automated aquaponic system that integrates aquaculture and agriculture. The system treats three parameters (water level, temperature, and light intensity), and contains four actuators, a DC fan, a heater, an LED, and pumps. Each parameter has been calibrated with a real device. The system has the order of (IF, then), exactly in the pump. For the bodies, the fish tank is 10 L, and the greenhouse is less than .05 m^2, which achieves the design requirements. During simulated testing, the prototype recorded an average temperature of 20.34°C, with a minimum of 18.31°C and a maximum of 36.75°C, showing generally stable conditions with occasional increases. The light intensity (lux) had a mean of 827.39 lux, ranging from a minimum of 31 lux to a maximum of 5884 lux, indicating wide variation in lighting levels. Additionally, the ultrasonic sensor measured distance with an average of 17.98 cm, a minimum of 5.20 cm, and a maximum of 39.98 cm, reflecting changes in water level during system operation. Overall, the prototype demonstrated the ability to monitor environmental parameters and respond automatically to maintain stable conditions suitable for aquaponic plant growth.

Recommendations

Predictive control systems are recommended for this project to anticipate deviations and trends in parameters like the fish tank's water level, water temperature, and the plant's light intensity before they occur. Predictive systems use the current process measurements, the current dynamic state of the process, and the process variable targets and limits to calculate future changes in the dependent variables. These changes are calculated to keep the dependent variables close to their targets while honoring constraints on both independent and dependent variables. Although a simplified rule-based approach is used instead of a full mathematical MPC

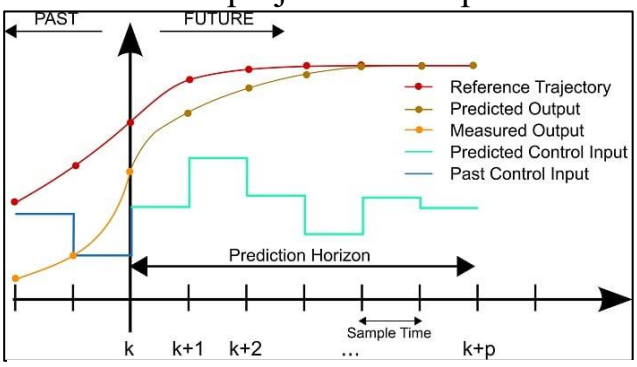


Figure 19 shows the importance of the predictive system.

model, the system maintains the core concept of predictive decision-making to enhance stability, efficiency, and resources**, as shown in Figure 19.**

<u>A Water generator</u>

A water generator is recommended to be implemented in this aquaponics system. This component relies on gravity and water flows from the upper plant pipe to the lower fish tank. This generator converts the mechanical energy into electric energy by letting water enter it and turn the turbine, and then it goes out. Some terms are needed to complete this process at full face: minimum height 2 meters, the pipe width should be doubled than ours. This addition will make this project more sustainable and decrease power consumption; as a result, it provides electricity for the microcontroller to work continuously. The generator **is shown in Figure 20**.

Figure 20 shows the water generator

Real-life application

We chose Borg El Arab City in Alexandria government to be our real-life location, **as shown in Figures 21 and 22**, due to its environmental and demographic characteristics that contrast with our school's location. Unlike the arid desert climate of October Gardens, Borg El Arab features a coastal environment with an average humidity of 70% and a moderate, stable climate year-round. This selection allows us to address the challenge of increasing the industrial and agricultural base of Egypt. Furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egyptian Grand Challenge of dealing with population growth and its consequences. The city provides a broad land availability for scalable agriculture, serving as a strategic hub to increase Egypt's agricultural base.

Figure 21 shows Borg El Arab

Figure 22 shows the beach of Borg El arab

Recommendations for Other Teams

As a capstone team, we will suggest some advice to another future team to improve their capstone. Firstly, the distribution of rules in an organized manner is critical to capstone teamwork. Second, avoid any tension between team members. That will affect your team negatively. Third, while making the prototype, you should choose the materials carefully before starting the prototype to avoid getting into trouble. Fourth, gain the experience from grade 11 or grade 12 to help you in a capstone. That will help you make posters, portfolios, and prototypes perfectly. In the end, avoid doing your tasks lately. This reason is the most important thing in the capstone.

Los table

LO	Math LO 1.02	PHY 2.03	Chem LO 1.01	Mech LO 2.01	GEO LO 2.01
Description	Using the type of graphs and the different ways to measure the error in sensors, such as MAE & MSE	DC-Circuits, connections in the circuits, voltage, and power consumption.	Determine the type of error during the calibration and the offset or coefficient	Calculating the mechanical power of the movement of the water in the system	Mechanism and the structure of the filtering of the water to evolve and create (BIO Filter)

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